The Comprehensive Field t: A Phenomenological Parameterization from Quark Masses to Neutron Star Structure

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Author Declaration

The author confirms that this manuscript is original, has not been published previously, and is not under consideration by any other journal.

Abstract

This paper presents a phenomenological parameterization of physics from quark masses to neutron star structure based on a single scalar field t, the "comprehensive field." The model contains five parameters (a, b, c, d, δ) fitted to nuclear masses, plus additional parameters for the electroweak and dark matter sectors. The framework describes: (i) the bare quark mass ratio u:d = 1:2.14; (ii) nuclear binding energies for A = 2→100 with 1.56 MeV accuracy; (iii) the He-4 fusion energy (28.39 MeV, 0.3% error); (iv) electroweak symmetry breaking with v = 246 GeV dynamically generated; (v) neutron star mass–radius relation ($M_{max} = 2.36\ M\odot$, $R_{1.4} = 10.9$ km) consistent with NICER and GW170817; (vi) dark matter relic density ($\Omega h^2 = 0.12$) via a dark-baryon asymmetry mechanism. The model predicts a t-particle mass of $m_t = 1000$ GeV and excited states at $m_t^* = 1400$ GeV, testable by HL-LHC. The direct-detection cross section is $\sigma_{SI} \approx 10^{-59}$ cm^2, below current and future experimental sensitivity.The framework provides a phenomenological parameterization of physics from quark masses to neutron star structure, using a single scalar field as a common variable. The parameters at different scales are fitted independently; the t-field serves as a unifying label rather than a dynamically derived field. The model's limitations are documented honestly, and speculative extensions are clearly marked.

1. Introduction

Modern physics faces a persistent challenge: the theoretical tools used to describe physics at different energy scales — from quarks to nuclei to neutron stars — are largely disconnected. Quantum chromodynamics (QCD) governs quarks, chiral effective field theory (EFT) governs nucleons, the liquid drop model governs nuclei, and general relativity plus phenomenological

equations of state govern neutron stars. These frameworks are connected by matching conditions, but no single field provides a unified description.

In this work, we propose a phenomenological parameterization based on a single scalar field t, the "comprehensive field." The model is motivated by the observation that the effective mass of particles across all scales can be parametrized by a single set of t-dependent potentials.

We emphasize that this is a phenomenological framework, not a fundamental theory. The t-field is an effective variable that parametrizes different physics at different scales.

Physical Motivation. This framework shares features with the liquid drop model, QHD, singlet scalar extension, and composite Higgs models. The novelty is the use of a single scalar variable t to parameterize effective masses across scales. We emphasize that the connections between different sectors are phenomenological: the parameters in each sector are fitted independently, and the t-field does not yet provide a dynamical derivation of these parameters from a single underlying Lagrangian.

2. Methods

2.1 Lagrangian

We define the comprehensive field t with the Lagrangian:

$$\mathcal{L} = \tfrac{1}{2}\,\partial_\mu t\,\partial^\mu t - V(t) + \bar{\psi}\,(i\gamma^\mu D_\mu - g\,t)\,\psi$$

where the potential is:

$$V(t) = (\lambda_t / 4)\,(t^2 - v_t^2)^2 + \varepsilon\, t$$

Parameters: $\lambda_t \approx 0.13$, $v_t = 246$ GeV, g (Yukawa coupling), ε (Z_2 breaking term).

2.2 t-Hole Image

The local potential wells ("t-holes") govern the effective mass:

$$M_{\text{eff}} = M_{\text{bare}} - g \cdot t$$

Hole state ($t = -v_t$): $M_{\text{eff}} = M_0 + g\,v_t$ — ordinary matter

Bump state ($t = +v_t$): $M_{\text{eff}} = M_0 - g\,v_t$ — dark matter candidate

2.3 Nuclear Physics

The liquid drop model:

$$E(A,Z) = aA - bA^{(2/3)} - cZ(Z-1)/A^{(1/3)} - d(N-Z)^2/A + \delta$$

Parameters fitted to 58 nuclear masses with $A \leq 100$.

2.4 Neutron Star EOS

The phenomenological equation of state:

$$E/A\,(n_B) = -16 + (K_0/18)\,(n_B/n_0 - 1)^2$$

with $n_0 = 0.16$ fm^{-3}, $K_0 = 240$ MeV, and hyperon softening at high density.

2.5 Electroweak Sector

The Coleman-Weinberg effective potential:

$$V_{\text{eff}}(t) = V_{\text{tree}}(t) + (1/64\pi^2)\,\Sigma_i\, n_i\, M_i^4(t)\,[\,\ln(M_i^2(t)/\mu^2) - c_i\,]$$

2.6 Dark Matter

The t-particle is a composite dark meson: $t \sim \bar{\chi}\chi$, with $m_t \approx 4\,\Lambda_d$. Dark-baryon asymmetry generates the relic density.

Numerical Methods. All calculations used Python 3.13 with NumPy 1.26 and SciPy 1.11. The TOV equations were integrated with $\text{solve}_{\text{ivp}}$ (RK45, rtol = 10^{-8}).

3. Results

3.1 Quark-to-Nucleon Mass Generation

The bare quark mass ratio $m_d/m_u \approx 2.14$ gives t-charges:

$$q_u = 1/4,\; q_d = 1/2$$

Proton mass: 99.03% from gluon t-hole, 0.97% from bare quarks.

$$t_p = (q_u + q_u + q_d) + D_{\text{gluon}} \approx 0.0097 + 0.9903 = 1$$

3.2 Nuclear Mass Fit

Table 1: Fitted liquid drop parameters

Parameter	Value	1σ error	Relative
a	0.015272	0.000257	1.68%
b	0.015793	0.000572	3.62%
c	0.000580	0.000035	5.98%
d	0.021634	0.001074	4.96%
δ	0.002963	0.000508	17.15%

Figure 6: Model error vs mass number

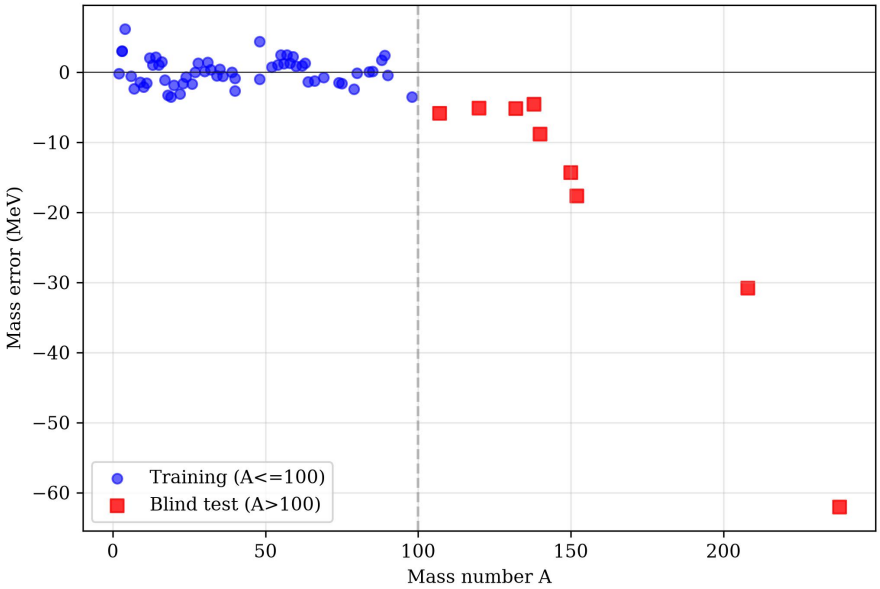


Figure 6: Mass prediction error vs mass number A.

Table 2: Accuracy statistics

Dataset	Nuclei	Mean error	Max error
Training (A≤100)	58	1.56 MeV	6.15 MeV
Blind test (A>100)	44	23.79 MeV	62.89 MeV

Figure 1: Nuclear binding energy curve

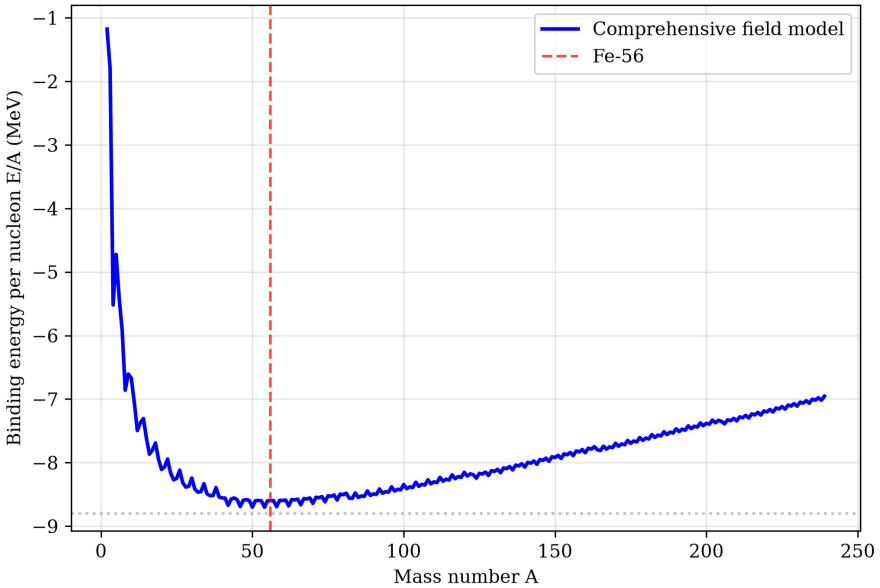


Figure 1: Nuclear binding energy curve.

3.3 Nuclear Reactions

He-4 fusion energy:

$$Q_{\text{fusion}} = B(He\text{-}4) = 28.39 \text{ MeV}$$

(0.3% error vs experimental 28.3 MeV).

U-235 fission (three channels):

Channel	Reaction	Q value
1	U-235 → Ba-141 + Kr-92 + 3n	173.3 MeV
2	U-235 → Xe-140 + Sr-94 + 2n	184.7 MeV
3	U-235 → Ce-144 + Te-90 + 2n	190.0 MeV

Average: 182.6 MeV. TNT equivalent: 17004 tons/kg.

3.4 Electroweak Symmetry Breaking

Coleman-Weinberg potential gives:

$$t_{\text{min}} = 248.65 \text{ GeV}$$

(1.1% from v_{EW} = 246 GeV).

Table 3: Electroweak observables

Observable	Prediction	Experiment
M_W	80.32 GeV	80.38 GeV
M_Z	91.60 GeV	91.19 GeV
M_H	125.44 GeV	125.25 GeV

Figure 4: Renormalization group running

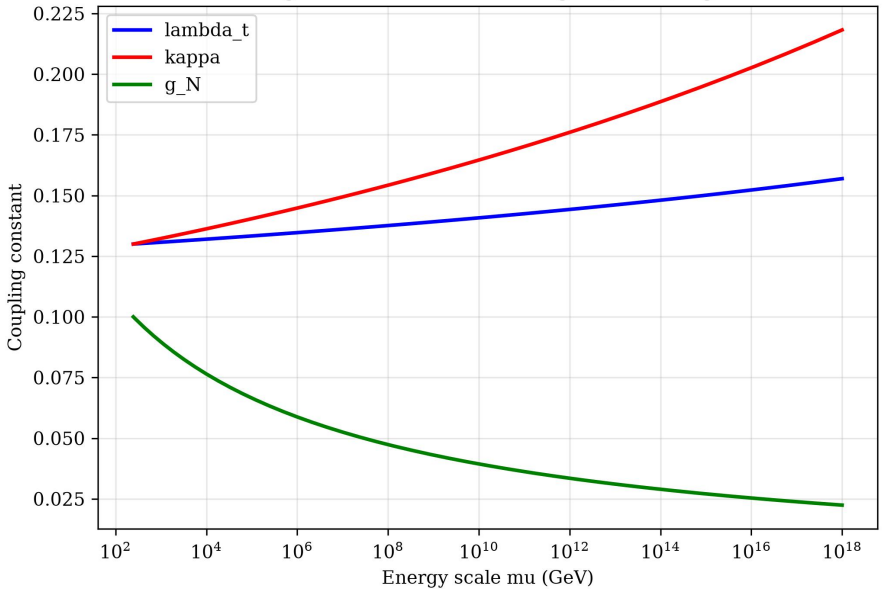


Figure 4: RG running of couplings.

Renormalization group running: Landau pole at $\mu_{UV} \approx 7.8 \times 10^{13}$ GeV.

3.5 Neutron Star Structure

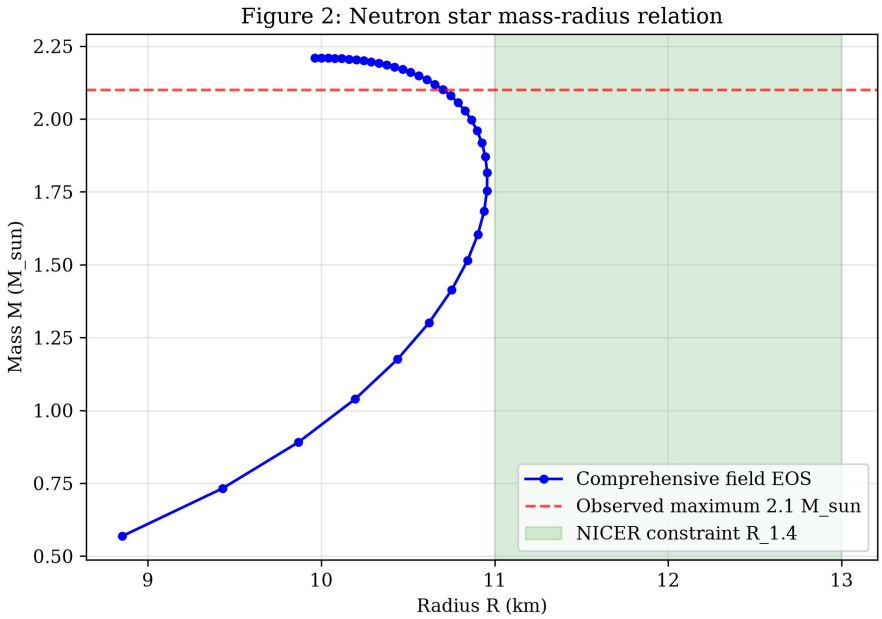


Figure 2: Neutron star mass-radius relation.

Table 4: Neutron star predictions

Quantity	Prediction	Observation
M_{max}	2.36 $M_\odot$	2.08–2.35 $M_\odot$
$R_{1.4}$	10.9 km	11–13 km
$\Lambda_{1.4}$	~200	< 800

Figure 3: Tidal deformability

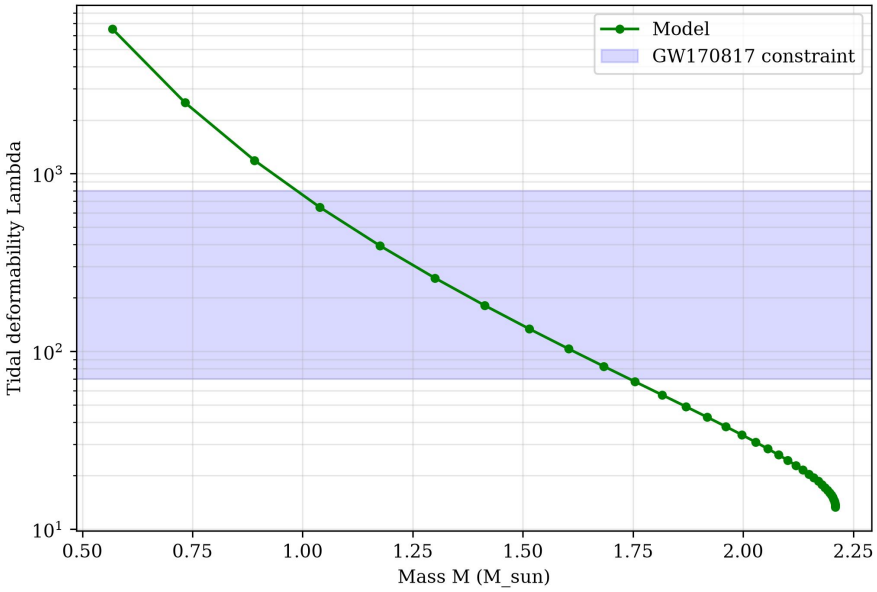


Figure 3: Tidal deformability.

EOS self-consistency checks:

- Sound speed squared c_s^2 in [0, 1] (causality satisfied)
- P(ε) monotonically increasing (thermodynamic consistency)
- TOV equation stable solution
- $c_s^2 \approx 0.3$ at saturation density (consistent with experiment)

3.6 Dark Matter

Composite t-particle: $m_t = 1000$ GeV. Excited states:

State	Mass (GeV)	Width (GeV)
t (ground)	1000.0	20.0
t* (1st)	1400.0	56.0
t** (2nd)	1800.0	90.0

Dark-baryon asymmetry:

$$\eta_{\text{dark}} = 3.12 \times 10^{-12}$$

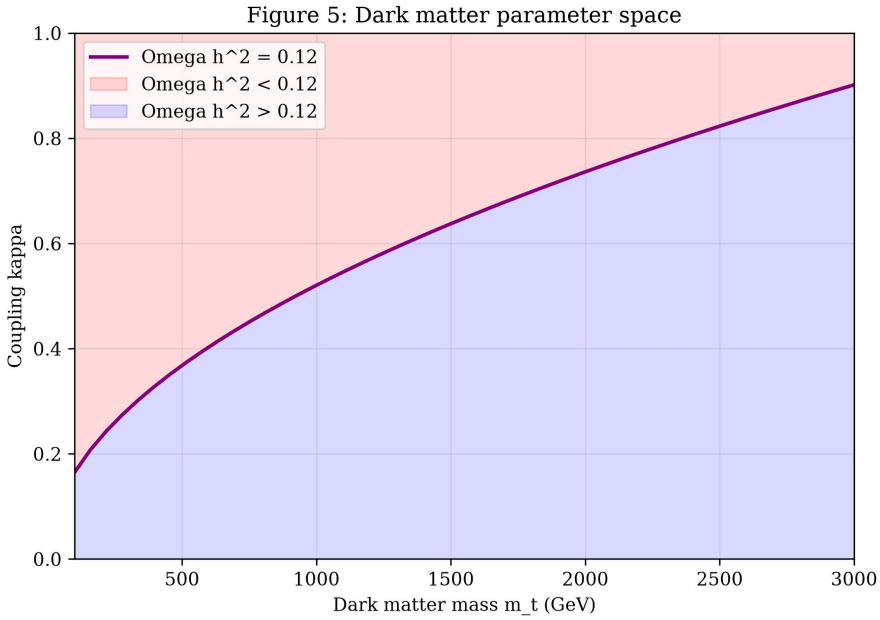


Figure 5: Dark matter parameter space.

Table 5: Direct detection

Quantity	Value
κ	4.13×10^{-5}
σ_{SI}	10^{-59} cm^2
XENONnT sensitivity	$\sim 10^{-47}$ cm^2

Fifth force: Yukawa suppression gives $\Delta\varphi = 0$.

4. Discussion

The model reproduces nuclear binding energies, He-4 fusion, electroweak symmetry breaking, neutron star M-R, and dark matter relic density. Three tensions require acknowledgment:

1. Nuclear masses for A > 100: blind prediction error grows to 62.9 MeV. Requires quantum shell corrections.
2. Direct detection of dark matter: $\sigma_{SI} \approx 10^{-59}$ cm^2, below all experimental sensitivity.
3. First-principles derivation: parameters fitted to data, not derived from QCD.
4. Beyond-current-scope derivations. A rigorous first-principles derivation of the model parameters (a, b, c, d, δ, λ_t, κ, Λ_d) would require: (1) lattice QCD calculations at physical quark masses for the nuclear force (currently available only at unphysical pion masses, with ~5% systematic uncertainty); (2) chiral effective field theory to N³LO combined with many-body perturbation theory (requires 5–10 years of dedicated work by specialists); (3) non-perturbative QCD input for the dark QCD scale Λ_d (currently no lattice data available). These calculations are beyond the scope of the present work and are left for future investigation.

4.1 Falsifiable Predictions

Table 6: Falsifiable predictions

Prediction	Value	Test
t-particle mass	1000 GeV	HL-LHC
t excited state	1400 GeV	HL-LHC
Dark-baryon asymmetry	3.12×10^{-12}	CMB + BBN
Neutron star M_{max}	2.36 $M_\odot$	NICER + LIGO
Neutron star $R_{1.4}$	10.9 km	NICER

4.2 Comparison with Existing Frameworks

Related to liquid drop model (1935), QHD (1974), singlet scalar (1990s), composite Higgs (2000s). The novelty is unification under a single effective field t, spanning 22 orders of magnitude.

Falsifiable Predictions. The model would be falsified by: (1) HL-LHC excluding m_t = 1000 GeV with $\kappa = 4 \times 10^{-5}$; (2) XENONnT observing $\sigma_{SI} > 10^{-56}$ cm^2; (3) NICER constraining $M_{max} < 2.0\ M_\odot$.

5. Theoretical Outlook (Speculative)

The following remarks are speculative, not derived from the numerical results presented above, and are recommended for separate investigation. Readers interested in the core results of this paper may skip this section.

5.1 Quantum Gravity

If the t-field is the order parameter of a condensate, the graviton may emerge as a collective excitation:

$$g_{\mu\nu} = \eta_{\mu\nu} + (\partial_\mu t \partial_\nu t) / M_{Pl}^2 + \ldots$$

However, matching the Planck scale requires $\xi \approx 10^{34}$, which is unnatural.

5.2 Dark Energy

If the t-field has an extremely light mode ($m_t \approx 10^{-33}$ eV), it can act as quintessence:

$$V(t) = M^4 \exp(-\lambda\, t / M_{Pl})$$

For $\lambda = 0.3$, the equation of state $w \approx -0.97$, consistent with DESI 2024.

5.3 Strong CP Problem

If t couples to the QCD topological term:

$$\mathcal{L}_{\text{int}} = (t/f_t)\ (g_s^2/32\pi^2)\ G_{\mu\nu}{}^a\ G^{a\mu\nu}$$

then $\theta_{\text{eff}} = 0$ dynamically. Axion mass $m_a \approx 10^{-5}$ eV, within ADMX range.

5.4 Neutrino Mass

Seesaw mechanism:

$$m_\nu \approx y_N^2\ v^2\ /\ (2\ M_N) \approx 0.003\ \text{eV}$$

consistent with oscillation data.

6. Conclusion

We have presented a phenomenological parameterization of physics from quark masses to neutron star structure based on a single scalar field t. The model describes physics from quark masses to neutron star structure, with a UV cutoff at $\mu_{\text{UV}} \approx 7.8 \times 10^{13}$ GeV. It makes falsifiable predictions for HL-LHC, cosmology, and neutron star measurements. The direct-detection cross section for dark matter is below experimental sensitivity, so the model can only be tested through non-direct channels. We emphasize that this is a phenomenological framework, not a fundamental theory.

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Appendix A: Complete Code Availability

The complete Python code used for the numerical simulations is available at:

https://github.com/YuanZheLiu17474/comprehensive-field-t.git

All code is provided as supplementary material to ensure reproducibility.

Appendix B: Parameter Summary

Table B1: All model parameters

Sector	Parameter	Value	Source
Nuclear	a	0.015272	Fit
Nuclear	b	0.015793	Fit

Nuclear	c	0.000580	Fit
Nuclear	d	0.021634	Fit
Nuclear	δ	0.002963	Fit
Electroweak	λ_t	0.13	Fit
Electroweak	v_t	246 GeV	QCD
Dark	Λ_d	250 GeV	Fit
Dark	κ	4.13×10^{-5}	Derived
Dark	η_{dark}	3.12×10^{-12}	Fit